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# Comparison of Computed Tomographic Features of Confirmed Nasal Neoplasia in Dogs

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## ABSTRACT

While computed tomography (CT) imaging is commonly used to evaluate canine nasal tumors, the ability to differentiate tumor types based on imaging features remains limited. This retrospective study examined dogs with confirmed nasal neoplasia to determine whether CT characteristics differ between epithelial and mesenchymal tumors. Cases from a single institution (2013–2022) were reviewed, and the frequency of CT features between epithelial and mesenchymal tumors was compared using Fisher's exact test. Also, CT features of less commonly reported nasal neoplasms, including squamous cell carcinomas, polyps, and osteosarcomas, were described. Sixty-seven dogs with nasal neoplasia were identified; 48 (72%) had a type of epithelial neoplasia, and 19 (28%) had mesenchymal neoplasia. Dogs with epithelial neoplasia were more likely to show intracranial mass extension ($p = 0.04$; OR 5.1; 95% CI 1.1–23.9), cribriform plate lysis ($p = 0.03$; OR 4.5; 95% CI 1.2–15.8), lysis of ipsilateral sphenoid sinus ($p < 0.0001$; OR 18.7; 95% CI 3.9–85.9), mass extension into ipsilateral sphenoid sinus ($p = 0.01$; OR 5.8; 95% CI 1.6–20.2), and frontal sinus fluid ($p = 0.05$; OR 4.7; 95% CI 1.3–16.2) than dogs with mesenchymal neoplasia. Dogs with mesenchymal neoplasia were more likely to show fluid in the ipsilateral maxillary recess ($p = 0.01$; OR 5.3; 95% CI 1.4–18.6). Squamous cell carcinoma patients had two distinct presentation patterns: either a small nodule centered on the nasal planum with no associated lysis and mass extension, or a mass centered on and causing lysis of the maxillary or nasal bone. This investigation provides the first comprehensive comparison of CT characteristics between different canine nasal tumor types, offering potential prebiopsy diagnostic indicators.

## 1 | Introduction

Primary tumors of the nasal cavity are uncommon in dogs, accounting for approximately 1%–2% of all canine neoplasia [1], with the most common tumor types being epithelial and mesenchymal in origin [2]. Histopathology is the gold standard for diagnosing nasal tumor types; however, nonrepresentative sampling during biopsy, such as sampling peritumoral inflammation or necrotic tissue, can lead to inaccurate diagnosis [3]. In dogs, the diagnostic success rate of rhinoscopy-assisted biopsy was 83% in a group of 149 dogs [4]. Repeat biopsy is frequently required for definitive diagnosis, with approximately one third of the histological diagnoses of neoplasia made on repeated biopsies in a study of 117 dogs with intranasal neoplasia [3]. Computed tomography (CT), paired with rhinoscopy, is a common diagnostic modality used in dogs with clinical signs related to the nasal cavity, and CT is considered the first-line imaging procedure in dogs with a suspected nasal tumor [5]. CT can guide clinicians to regions most likely to provide a diagnosis and help avoid areas that may represent necrosis. Specific CT features of differing tumor types could aid in diagnosis when histopathology is inconclusive.

**Abbreviations:** CT, computed tomography; HU, Hounsfield units.

**TABLE 1** | CT features characterized for each patient.

| Feature | Findings recorded |
|---|---|
| Laterality of mass | Right, left, bilateral |
| Location mass is centered | Rostral, middle, caudal |
| Mass degree of contrast enhancement | Minimally, mildly, moderately, strongly |
| Mass pattern of contrast enhancement | Homogeneous, heterogeneous, peripherally |
| Presence of new mineral in mass | None, mild, moderate, severe |
| Presence, degree, and location of lysis | None, minimal, mild, moderate severe, all; nasal conchal (left, right, bilateral), ethmoidal (left, right, bilateral), vomer/septal, cribriform plate, ipsilateral orbit, nasal bone, maxillary bone, hard palate |
| Presence, subjective degree, and location of mass extension | None, minimal, mild, moderate, severe; Intracranial, nasopharyngeal, retrobulbar, intraoral, paranasal soft tissues |
| Expansion of maxilla | None, minimal, mild, moderate, severe |
| Periosteal reaction | None, minimal, mild, moderate, severe |
| Involvement of frontal sinuses | Presence, laterality and degree (none, minimal, mild, moderate, severe, all) of the following features: presence of fluid and/or mass extension (and if gravity dependent/independent), lysis of sinus margins |
| Involvement of ipsilateral maxillary recess | Presence and subjective degree (none, minimal, mild, moderate, severe, all) of the following features: lysis, mass extension, fluid |
| Involvement of ipsilateral sphenoid sinus | Presence and subjective degree (none, minimal, mild, moderate, severe, all) of the following features: lysis, mass extension, fluid |
| Presence, subjective degree, and location of fluid in each nasal cavity (left and right) | None, mild, moderate, severe, all; rostral middle, caudal, all |
| Regional lymph node size | Mandibular and medial retropharyngeal lymph nodes measured in millimeters in three dimensions |
| Regional lymph node hilus visibility | Yes, no |
| Regional lymph node contrast enhancement | Normal, abnormal |

A report describing the effectiveness of CT in differentiating between different causes of chronic nasal disease in dogs [6] concluded that CT has an accuracy greater than 90% for all disease processes evaluated (neoplasia, aspergillosis, nonspecific rhinitis, foreign body rhinitis). Another report evaluating the ability of CT to differentiate nasal neoplasia from nonneoplastic diseases (specifically between oronasal fistulas, lymphoplasmacytic rhinitis, purulent rhinitis, and unknown etiologies) in a group of 78 dogs with nasal symptoms [7] found that lysis of the nasal septum, hard palate, cribriform plate, and frontal sinus and invasion into the orbit and surrounding subcutaneous tissues were significantly more common in the neoplastic group.

Reports describing differences in CT characteristics among different nasal tumor types have been published in cats [8], supporting the ability of CT features to aid in the discrimination of tumor type (specifically lymphoma, non-lymphoma neoplasia, and inflammatory lesions). To the authors' knowledge, no published study exists in veterinary literature describing differences in CT features between types of nasal neoplasia in dogs.

The objectives of this study are (1) to compare CT features of confirmed epithelial and mesenchymal nasal tumors in dogs and (2) to determine if distinctive CT features exist between these different tumor types. Additionally, it aims to describe CT features of uncommon nasal tumors, including squamous cell carcinomas, nasal polyps, and osteosarcomas, which have limited representation in the literature. Our hypothesis is that CT features will differ between epithelial and mesenchymal nasal tumors—more specifically, lesion mineralization will be more common with mesenchymal tumors, particularly osteosarcoma.

## 2 | Materials and Methods

### 2.1 | Case Selection and Clinical Data Recording

The study design was a retrospective, single-center, cross-sectional study. Pet owners and the hospital director approved the use of the data. Medical records of the Carlson College of Veterinary Medicine Veterinary Teaching Hospital at Oregon State University (Corvallis, USA) were searched for dogs diagnosed with a mass of the nasal cavity between August 2013 and January 2022. Decisions for subject inclusion were made by a first-year diagnostic imaging resident (K.V.) and an American College of Veterinary Internal Medicine-certified small animal medicine internist (S.S.). Cases were included if the dog had a mass arising from the nasal cavity identified on head CT and a definitive diagnosis of benign or malignant neoplasia on histopathology of nasal tissue or cytology of a draining lymph node. A mass was defined as a soft-tissue, round-to-ovoid structure that caused regional destruction of the nasal conchae and/or displacement

**TABLE 2** | Summary of neoplasia types.

| Tumor type | Number of cases |
|---|---|
| Epithelial neoplasia group | 48 |
| Adenocarcinoma | 25 |
| Undifferentiated carcinoma | 14 |
| Squamous cell carcinoma | 5 |
| Polyp | 3 |
| Transitional cell carcinoma | 1 |
| Mesenchymal neoplasia group | 19 |
| Undifferentiated sarcoma | 7 |
| Osteosarcoma | 5 |
| Chondrosarcoma | 3 |
| Fibrosarcoma | 2 |
| Intranasal rhabdomyosarcoma | 1 |
| Malignant peripheral nerve sheath tumor | 1 |

of regional structures. Cases were excluded if the mass did not originate in the nasal cavity, frontal sinuses, or paranasal bones (e.g., oral cavity, skin) or if histopathology/cytology was unable to differentiate neoplasm type. In the case of repeat CT exams, only the initial study was included. For all dogs included, the following clinical data were recorded from the medical record by a single author (K.V.): signalment, presenting complaint, whether regional lymph node cytology was performed (yes with cytologic diagnosis/no), whether rhinoscopy was performed (yes/no), biopsy technique (blind vs. rhinoscopy assisted), and final histopathologic diagnosis of the nasal mass.

## 2.2 | CT Image Acquisition

Dogs were placed in ventral recumbency either under sedation or general anesthesia. Precontrast and postcontrast CT images of the head were acquired using a helical CT scanner (Toshiba Aquilion 64 slice; Tokyo, Japan). Postcontrast images were obtained 1 min after injection of a nonionic iodinated contrast medium (30% organically bound iodine [300 mg/mL], Isovue-300, Bracco Diagnostics, Milan, Italy) administered as an intravenous bolus (1–2 mL/kg) given over 65 s using a power injector (Empower CTA Injector System, Bracco Diagnostics, Milan, Italy). Digital images were reconstructed using high- and low-frequency algorithms, viewed in bone (window width [WW] = 2700; window level [WL] = 350) and soft tissue (WW = 320; WL = 30) windows, and available in transverse, sagittal, and dorsal planes. Acquisition parameters for CT images were as follows: 512 × 512 matrix, 2 mm slice thickness, 120 kVp, 250 mA, and 0.75 pitch for small dogs, and 400 mA and 0.6 pitch for large dogs.

## 2.3 | CT Image Analysis

CT images were anonymized by a single author (K.V.) and uploaded to a web-based PACS system (Novarad, American

**TABLE 3** | Abbreviated table of CT findings for epithelial neoplasia and mesenchymal neoplasia groups (for full table, see Supporting Information).

| Feature | Epithelial neoplasia ($n = 48$) | Mesenchymal neoplasia ($n = 19$) |
|---|---|---|
| Laterality of mass | | |
| Unilateral | 41 (85.4%) | 17 (89.5%) |
| Bilateral | 7 (14.6%) | 2 (10.5%) |
| Location mass centered | | |
| Rostral | 4 (8.3%) | 1 (5.3%) |
| Mid | 22 (45.8%) | 13 (68.4%) |
| Caudal | 22 (45.8%) | 5 (26.3%) |
| Presence of new mineral in mass | 16 (33.3%) | 9 (47.4%) |
| Cribriform plate lysis | 22 (45.8%) | 3 (15.8%) |
| Ipsilateral sphenoid sinus lysis | 33 (68.8%) | 2 (10.5%) |
| Ipsilateral sphenoid sinus extension | 25 (52.0%) | 3 (15.8%) |
| Intracranial extension | 18 (37.5%) | 2 (10.5%) |
| Retrobulbar extension | 20 (41.2%) | 5 (26.3%) |
| Intraoral extension | 9 (18.8%) | 4 (21.5%) |
| Nasopharyngeal extension | 39 (81.3%) | 16 (84.2%) |
| Periosteal reaction | 13 (27.1%) | 6 (31.6%) |
| Fluid in frontal sinus | 41 (85.4%) | 13 (68.4%) |
| Fluid in ipsilateral maxillary recess | 24 (50%) | 16 (84.2%) |
| Enhancement of regional lymph nodes | | |
| All LN normally enhancing | 37 (77.1%) | 14 (73.7%) |
| At least one abnormally enhancing LN | 11 (22.9%) | 5 (26.3%) |
| Presence of regional lymph node fat hilus | | |
| Present in all LN | 29 (60%) | 17 (77.3%) |
| At least one LN with absent fat hilus | 19 (40%) | 5 (22.7%) |

Fork, UT, USA). Each case was independently reviewed, and feature data were independently collected by two ACVR-certified veterinary radiologists (S.C. and L.N.) with 8 and 5 years of experience, respectively. With all features evaluated, in the case of nonagreement between radiologists, the images were jointly re-evaluated, and a consensus was reached. The radiologists were not restricted in plane selection, ability to window/level, or zoom/pan images as required for evaluation. The radiologists were aware of the diagnosis of nasal neoplasia at the time of image interpretation; however, they were blinded to the final histopathologic diagnosis and did not access or use the official radiology reports at any point during the study.

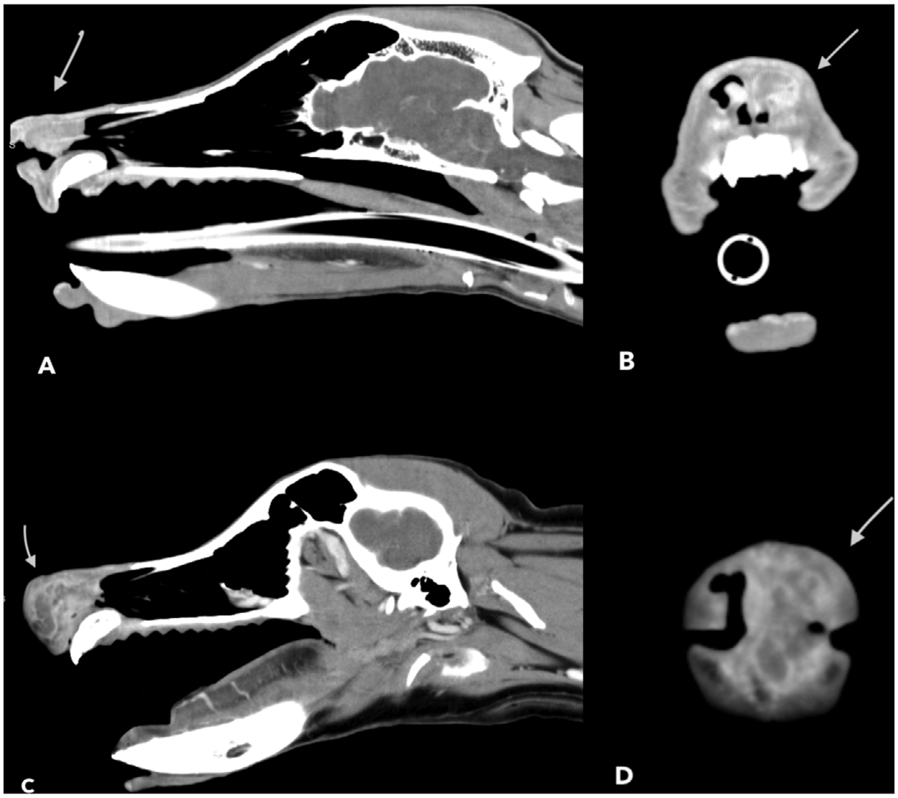


**FIGURE 1** | Contrast-enhanced computed tomographic images with a soft tissue algorithm (window width = 320 HU, window level = 30 HU, slice thickness = 2 mm) of two dogs diagnosed with nasal planum squamous cell carcinoma. (A) Para-sagittal and (B) transverse images of a dog demonstrating a contrast-enhancing nodule (white arrows) of the left aspect of the nasal planum, extending into the rostral nasal cavity. (C) Para-sagittal and (D) transverse images of a dog demonstrating a mildly larger contrast-enhancing nodule (white arrows) of the left aspect of the nasal planum, extending into the rostral nasal cavity.

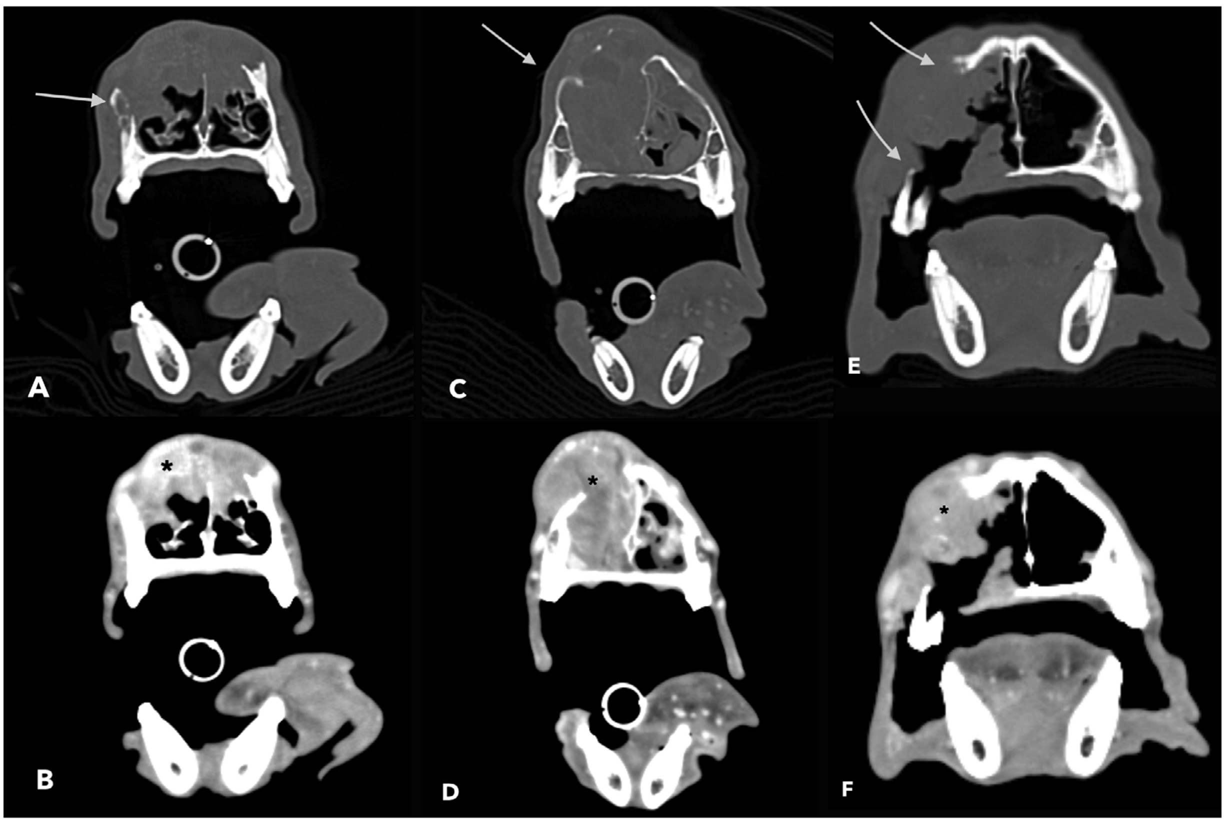


**FIGURE 2** | Legend on next page.

Multiple CT features were recorded for each case by each radiologist and are summarized in Table 1. These features include the location where the mass was centered (unilateral [right, left] or bilateral; rostral, middle, or caudal). Additionally, the nasal cavity was divided into six regions (right rostral, middle, and caudal and left rostral, middle, and caudal), and the approximate volume of the mass occupied in each region was recorded (none, minimal, mild, moderate, severe, or all). The term rostral was used to describe lesions rostral to the root tips of the maxillary canine teeth. The term middle was used to describe lesions between the root tips of the maxillary canines and the maxillary fourth premolar tooth (carnassial tooth). The term caudal was used to describe lesions caudal to the maxillary fourth premolar tooth, including the ethmoid conchae [9]. For approximate volume of each nasal region occupied by the mass, none was used when the mass did not occupy the region, minimal when approximately 5%–10% was occupied, mild when approximately 11%–33% was occupied, moderate when approximately 34%–66% was occupied, severe when greater than approximately 66% but less than 100% was occupied, and all when 100% was occupied.

Using the same definitions stated above, the presence, location (rostral, middle, caudal, all), and amount (none, minimal, mild, moderate, severe, all) of abnormal, non-mass fluid in the nasal cavity were recorded. Some cases had fluid in multiple locations and were recorded as such. Fluid was defined as nonenhancing soft tissue with an approximate HU of 0–20. Additionally, fluid in the ipsilateral sphenoid sinus and maxillary recesses was recorded (none, minimal, mild, moderate, severe, all).

The frontal sinuses were evaluated, and the presence, location, and amount of any soft tissue attenuation in either frontal sinus was recorded (none, minimal when 5%–10% was filled, mild when 11%–33% was filled, moderate when 34%–66% was filled, severe when greater than 66% but less than 100% was filled, and all when 100% of frontal sinus was filled). In the case where both frontal sinuses were affected with a different severity, the patient was assigned the most severe category present. Additionally, using the prior stated definitions, whether the soft tissue was a mass or fluid, and if the soft tissue was gravity dependent or independent, was recorded.

**FIGURE 2** | Transverse, contrast-enhanced, computed tomographic images of three dogs diagnosed with paranasal bone squamous cell carcinoma. All images demonstrate contrast-enhancing soft tissue tumors (black asterisks). The dog in images A and B has a tumor centered on the right maxillary and nasal bones; C and D show a different dog with a tumor centered on the right maxillary and nasal bones; and E and F show a third dog with a tumor centered on the right maxillary bone. All cases have severe lysis of these structures (white arrows) and mass extension into the paranasal soft tissues. The dog in E/F had severe lysis of the right hard palate and intraoral extension. Computed tomographic images were reconstructed with either a bone (A, C, E) or a soft tissue algorithm (B, D, F). Bone images are displayed with a window width of 2700 HU and a window level of 350 HU; soft tissue images are displayed with a window width of 320 HU and a window level of 30 HU. All images have a slice thickness of 2 mm, except E and F, which have 3 mm.

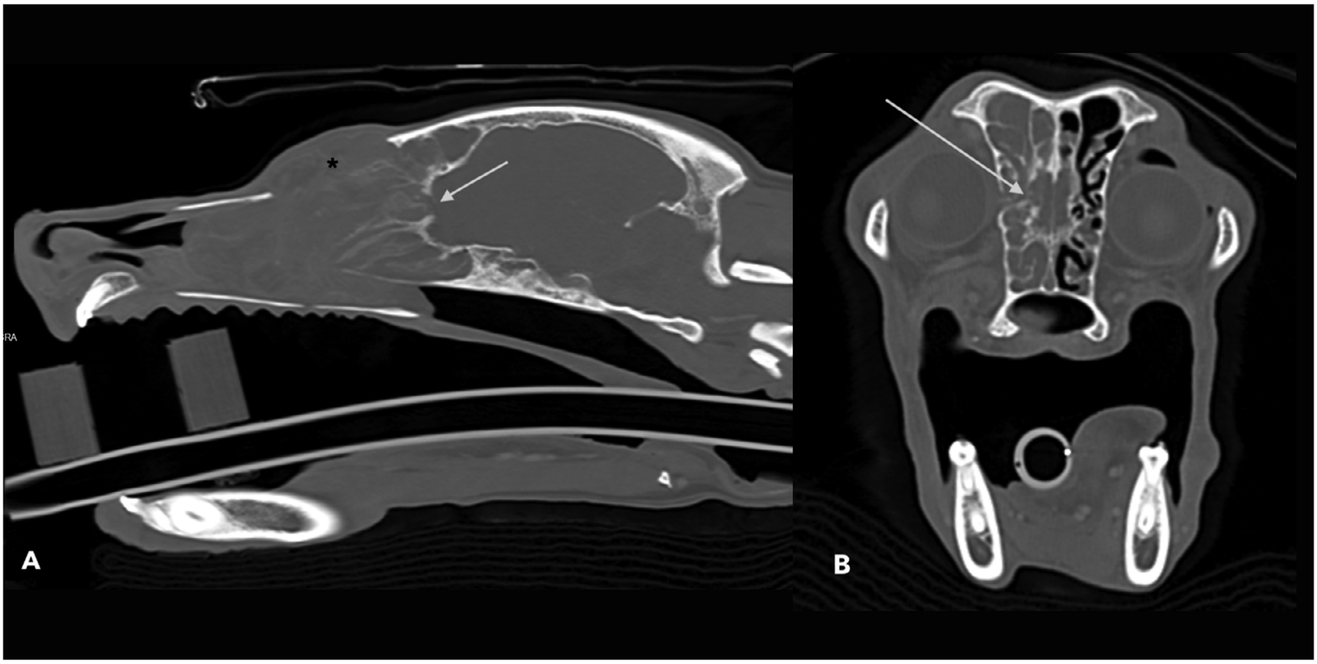


**FIGURE 3** | Sagittal (A) and transverse (B) contrast-enhanced computed tomographic images of one of the dogs diagnosed with nasal cavity squamous cell carcinoma (same patient as C and D in Figure 2). The tumor was centered on the right maxillary bone and nasal bones (black asterisk) with moderate nasopharyngeal mass extension, mild cribriform plate lysis (white arrow), and minimal intracranial mass extension. Computed tomographic images were reconstructed using a bone algorithm with a 2-mm slice thickness and are displayed with a window width of 2700 HU and a window level of 350 HU.

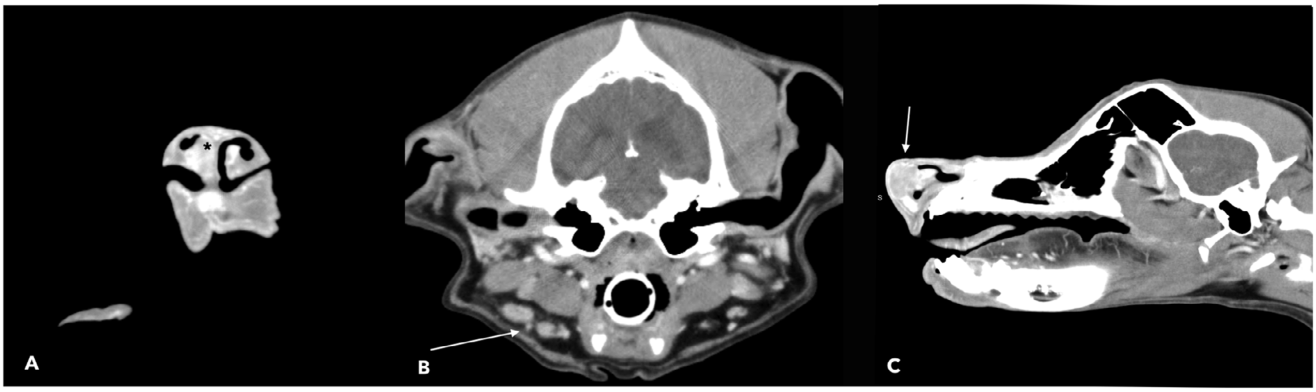


**FIGURE 4** | Contrast-enhanced computed tomographic images with a soft tissue algorithm (window width = 320 HU, window level = 30 HU, slice thickness = 2 mm) of a dog diagnosed with adenosquamous carcinoma of the nasal planum. Image A is a transverse and image C is a sagittal image demonstrating the small contrast-enhancing nodule of the right nasal planum (black asterisk and white arrow). Image B (white arrow) demonstrates the mildly enlarged right mandibular lymph nodes with adjacent mild fat stranding. Cytology of this lymph node revealed metastatic carcinoma.

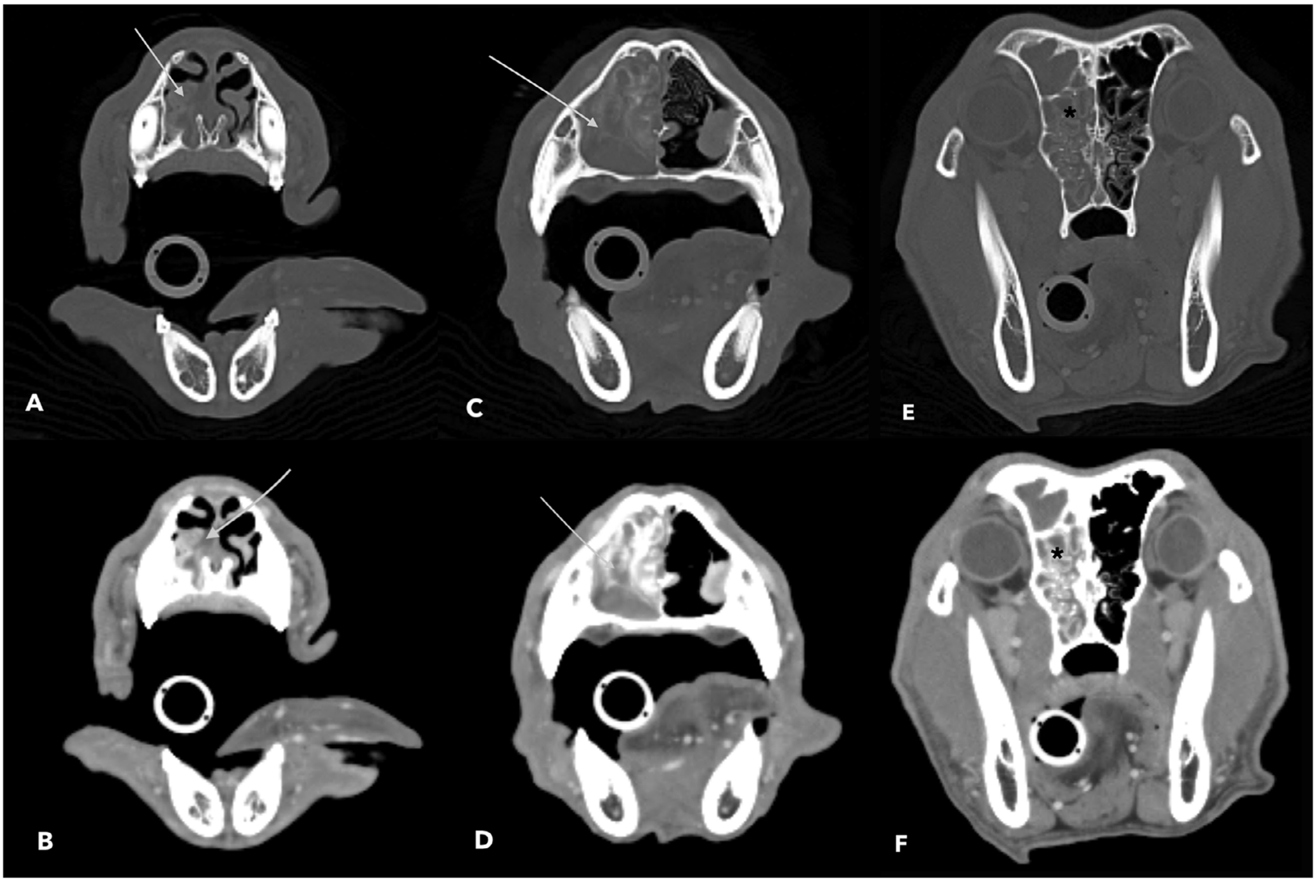


**FIGURE 5** | Transverse, contrast-enhanced, computed tomographic images of a dog diagnosed with osteosarcoma in the nasal cavity. Bone (A, C, E) and soft tissue images (B, D, F) of the rostral (A and B), mid (C and D), and caudal (E and F) aspects of the nasal cavity show a moderately contrast-enhancing mass centered in the right mid nasal cavity with severe new mineralization (white arrow in C), causing moderate lysis of the nasal conchae (white arrows D) and mild nasal septal lysis with mild rostral extension (white arrows A and B). Images E and F (black asterisks) show fluid in caudal nasal cavity. Bone images are displayed with a window width of 2700 HU and a window level of 350 HU; soft tissue images are displayed with a window width of 320 HU and a window level of 30 HU. Slice thickness 2 mm.

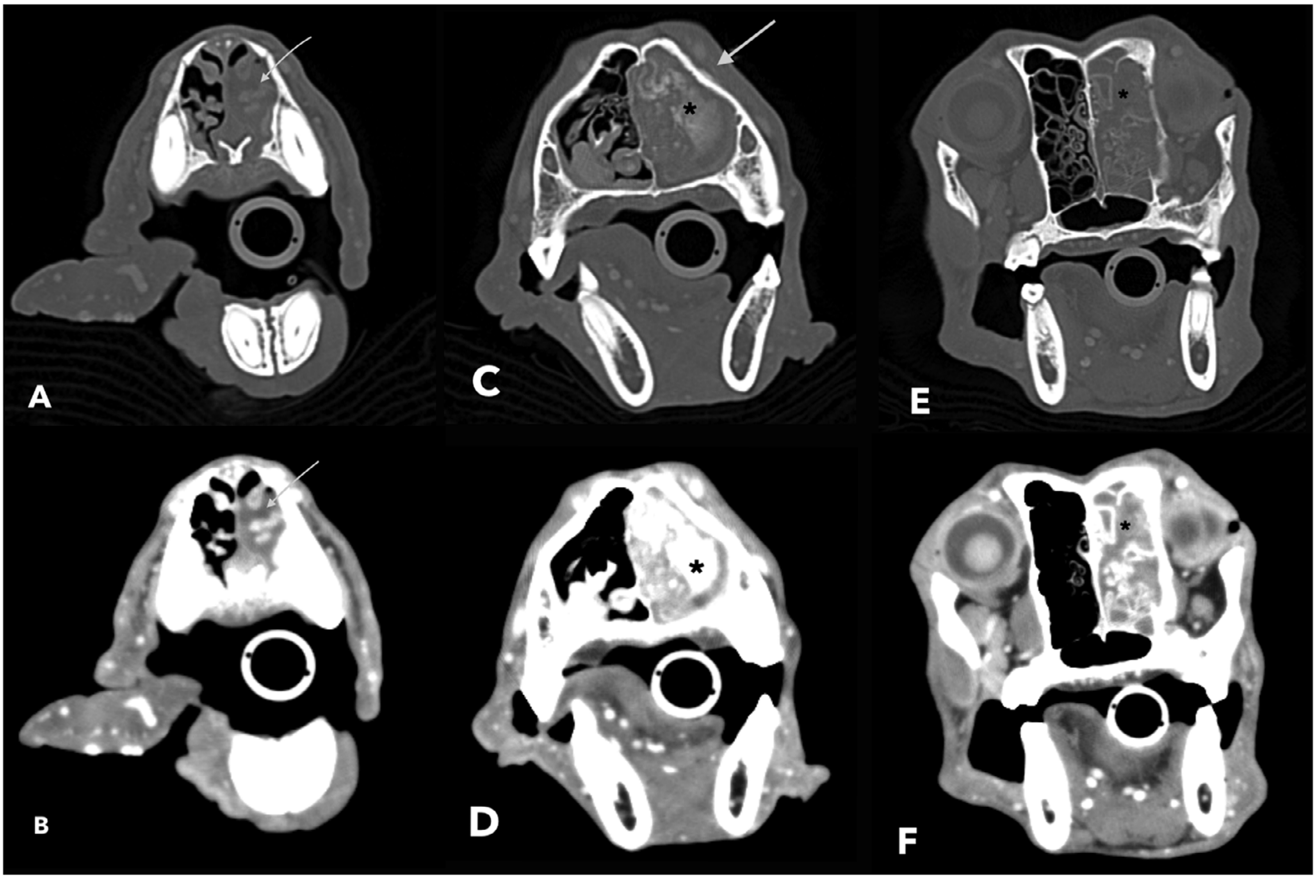


**FIGURE 6** | Transverse, contrast-enhanced, computed tomographic images of a dog diagnosed with osteosarcoma in the nasal cavity. Bone (A, C, E) and soft tissue images (B, D, F) of the rostral (A and B), mid (C and D), and caudal (E and F) aspects of the nasal cavity show a moderately contrast-enhancing mass centered in the left mid nasal cavity, causing moderate lysis of nasal conchae with severe new mineralization (black asterisk A, C) and mild periosteal reaction (white arrow C). The mass extends caudally, causing lysis of the ethmoid conchae (E and F black asterisks). Images A and B (white arrows) show fluid in the rostral nasal cavity. Bone images are displayed with a window width of 2700 HU and a window level of 350 HU; soft tissue images are displayed with a window width of 320 HU and a window level of 30 HU. Slice thickness 2 mm.

The presence and subjective degree of periosteal reaction surrounding the mass (none, minimal, mild, moderate, severe), presence and subjective degree of new mineralization in the mass (none, mild, moderate, severe), and subjective degree (none, minimally, mildly, moderately, strongly) and pattern of contrast enhancement (none, homogeneous, heterogeneous [mildly, moderately, severely], peripheral) were recorded. New mineral in the mass was defined as a mineral (evaluated precontrast) not normally present in the nasal cavity. The presence, location (right vs. left), and degree of nasal conchal and ethmoidal conchal lysis (none, minimal, mild, moderate, severe) were also recorded. When describing lysis of the ethmoid and nasal conchae, the term "none" was used when there was no detectable lysis, "mild" when approximately one third of the structures were lytic, moderate when up to two thirds were lytic, and severe when more than two thirds were lytic. At the extreme ends of the spectrum, the radiologists were allowed to use the terms "minimal" when there was very little lysis, but subjectively much less than one third (such as with minor extension lateral of midline), and "all" when none of the listed structures was detectable. The presence and subjective degree of lysis (none, minimal, mild, moderate, severe, all; right or left if applicable) of the cribriform plate, orbit, maxilla (non-conchal), nasal septum, nasal bone, hard palate, and margins of the maxillary, sphenoid, and frontal sinus were recorded.

Additionally, the presence and subjective degree (none, minimal, mild, moderate, severe) of expansion of the maxillary bone and extension of the mass into surrounding regions (paranasal soft tissue, nasopharyngeal, retrobulbar, intraoral, and intracranial) were recorded. Nasopharyngeal, intraoral, retrobulbar, and intracranial extension were defined as any portion of the mass extending into these cavities. Paranasal soft tissue extension was defined as extension of the mass into the skin or subcutaneous tissues dorsal and/or lateral to the nasal cavity.

The maximal dorsoventral dimension (in transverse plane) of both the medial and lateral mandibular lymph nodes, and the maximal mediolateral dimension (in transverse plane) of the medial retropharyngeal lymph nodes were recorded for each side. Additional features recorded for each lymph node were fat hilus (present/absent) and contrast enhancement after intravenous contrast medium (normal/abnormal). Normal lymph node contrast enhancement was defined as homogeneous enhancement, and abnormal was defined as heterogeneous or ring enhancement.

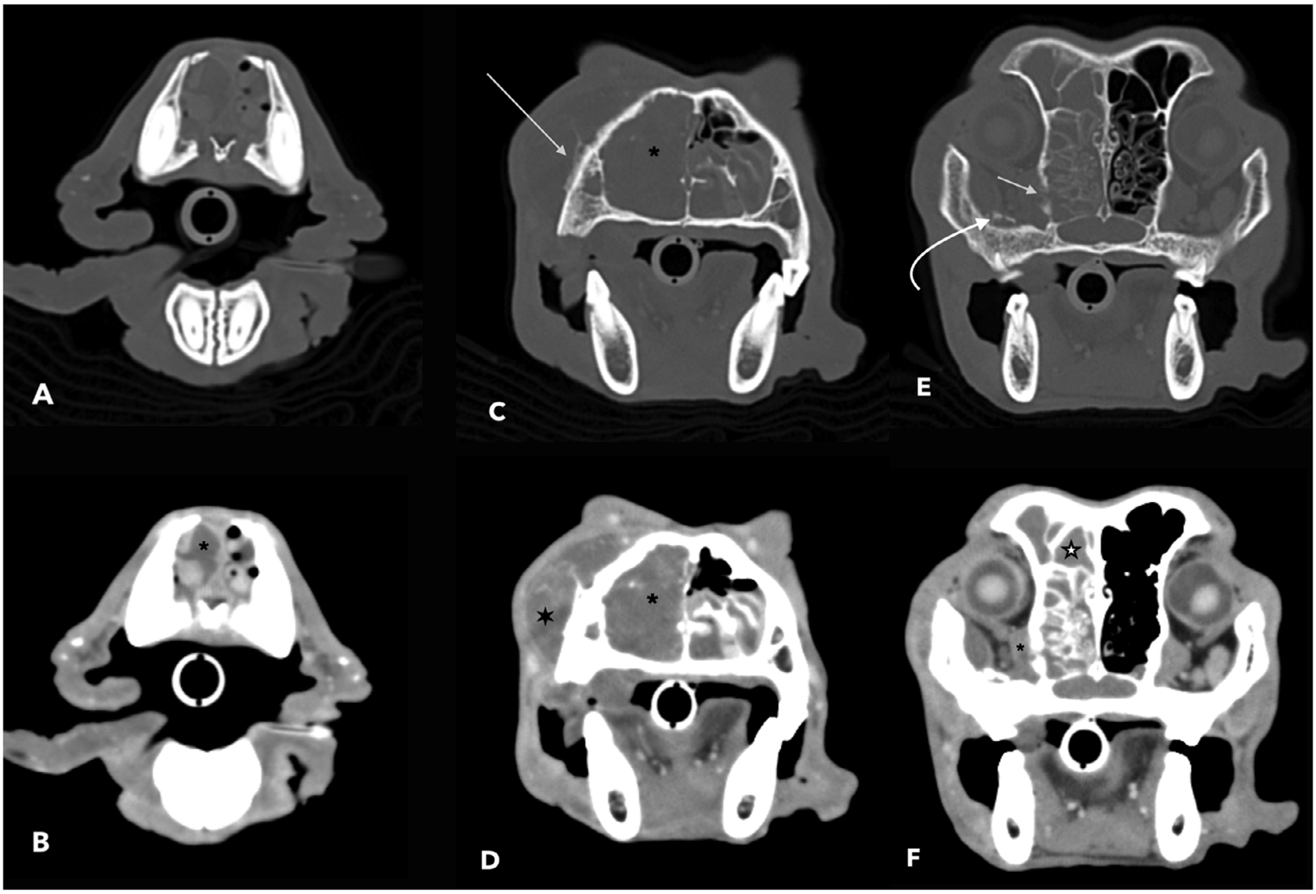


**FIGURE 7** | Transverse, contrast-enhanced, computed tomographic images of a dog diagnosed with osteosarcoma in the nasal cavity. Bone (A, C, E) and soft tissue images (B, D, F) of the rostral (A and B), mid (C and D), and caudal (E and F) aspects of the nasal cavity show mildly contrast-enhancing mass centered in the right mid nasal cavity, causing severe lysis of the nasal conchae (black asterisks in C and D), moderate paranasal soft tissue extension (large black star D), and severe periosteal reaction (white arrow in C, curved white arrow E). The mass extends caudally (large white star in F), and there was nonenhancing fluid rostrally (black asterisk B). There was mild lysis of the ipsilateral orbit (white arrow in E) and mild retrobulbar extension (black asterisk in F). Bone images are displayed with a window width of 2700 HU and a window level of 350 HU; soft tissue images are displayed with a window width of 320 HU and a window level of 30 HU. Slice thickness 2 mm.

## 2.4 | Statistical Analysis

Statistical analysis was performed by an American College of Veterinary Internal Medicine-certified small animal medicine internist (S.S.) with a background in statistics, using GraphPad Prism 10.2.2. A power analysis was not performed as the population was opportunistic. Dogs with epithelial neoplasia and those with mesenchymal neoplasia were compared. The following CT findings were categorized in binomial data and compared by using Fisher's exact test laterality (unilateral vs. bilateral): degree of contrast enhancement (none/minimal vs. mild to severe); pattern of contrast enhancement (homogeneous vs. heterogeneous); presence of mass extension (nasopharyngeal, intracranial, retrobulbar, paranasal soft tissue, intraoral); mass mineralization; bone lysis (conchal, ethmoidal, nasal septum, maxilla, cribriform plate, hard palate, nasal bone, ipsilateral orbit); expansion of maxilla; periosteal reaction; fluid in the nasal cavity and frontal sinuses (none/minimal vs. mild-severe); lysis, mass extension, and fluid in either the ipsilateral sphenoid sinus or maxillary recess; enhancement of regional lymph nodes (normal vs. at least one abnormally enhancing lymph node); and presence of regional lymph node fat hilus (present or absent in at least one lymph node). A $p$-value of $\leq 0.05$ was considered significant.

## 3 | Results

Sixty-seven dogs met the inclusion criteria. Sixty-six dogs had a diagnosis via histopathology of nasal tissue, and one dog had a diagnosis via cytology of the medial retropharyngeal lymph node (squamous cell carcinoma). Forty-eight dogs (72%) had a type of epithelial neoplasia—three had benign polyps (5%) and 45 (67%) had a type of carcinoma (25 adenocarcinomas, 14 undifferentiated carcinomas, five squamous cell carcinomas, and one transitional cell carcinoma)—and nineteen dogs (28%) had a mesenchymal neoplasia (seven undifferentiated sarcomas, five osteosarcomas, three chondrosarcomas, two fibrosarcomas, one malignant peripheral nerve sheath tumor, and one intranasal rhabdomyosarcoma). The number of patients diagnosed with each tumor is summarized in Table 2.

There were 26 mixed breed dogs, 11 Labrador Retrievers, four Golden Retrievers, two Beagles, three Poodles, and one of each

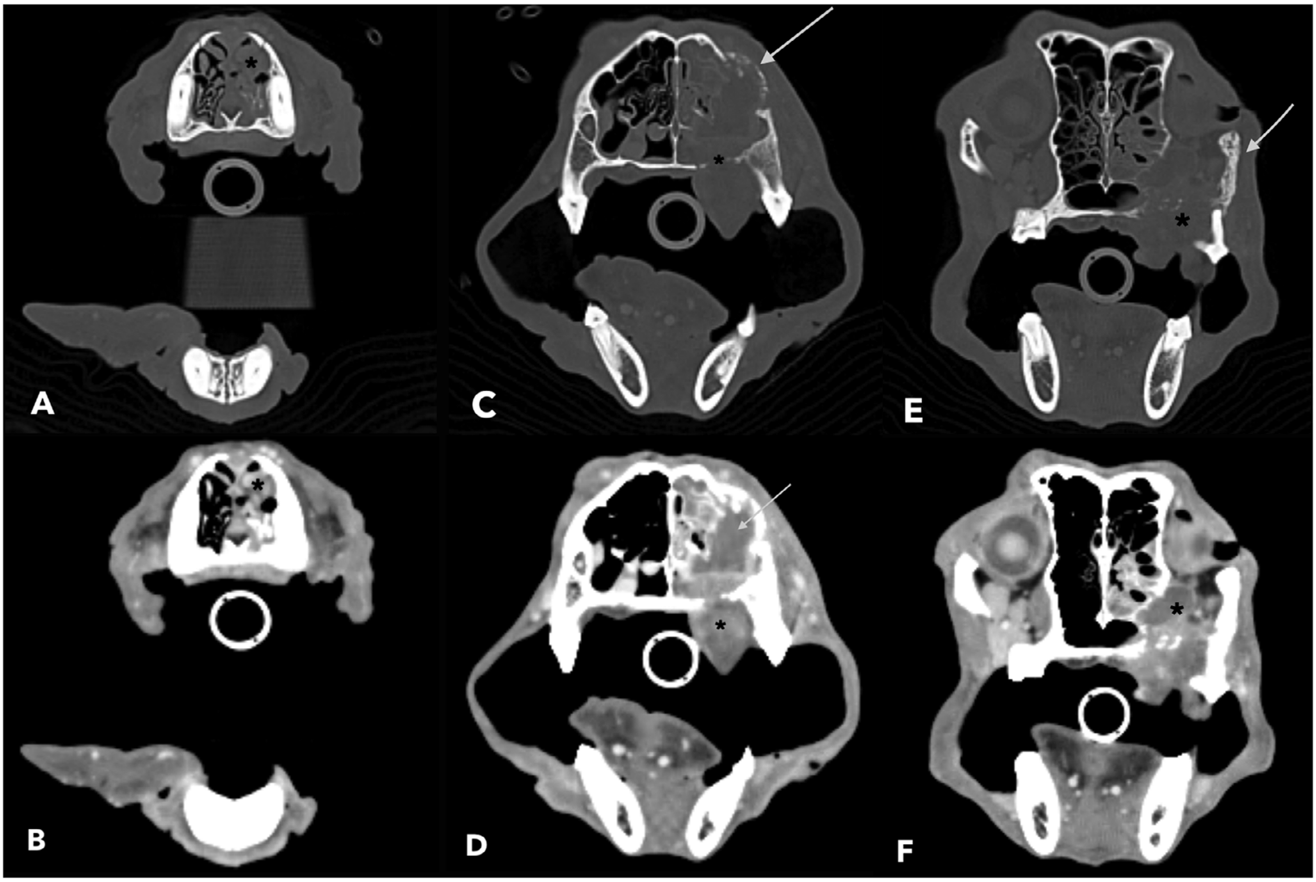


**FIGURE 8** | Transverse, contrast-enhanced, computed tomographic images of a dog diagnosed with osteosarcoma in the nasal cavity. Bone (A, C, E) and soft tissue images (B, D, F) of the rostral (A and B), mid (C and D), and caudal (E and F) aspects of the nasal cavity show a mildly contrast-enhancing mass centered in the left caudal nasal cavity, causing moderate lysis of the conchae (white arrow in D), severe lysis of the hard palate (black asterisk in C and E), moderate lysis of the zygomatic process of the maxilla (white arrow E), severe intraoral extension (black asterisk in D), and severe lysis of the ipsilateral orbit with moderate retrobulbar extension (black asterisk in F). The rostral nasal cavity had mild fluid (black asterisk A and B). Bone images are displayed with a window width of 2700 HU and a window level of 350 HU; soft tissue images are displayed with a window width of 320 HU and a window level of 30 HU. Slice thickness 2 mm.

of the following breeds: Chihuahua, Newfoundland, Dachshund, American Bulldog, Springer Spaniel, Irish Setter, Nova Scotia Duck Tolling Retriever, Labradoodle, Border Terrier, Doberman Pinscher, Alaskan Klee Kai, Yorkshire Terrier, English Pointer, Bernese Mountain Dog, Belgian Tervuren Dog, Border Collie, Australian Shepherd, Flat Coated Retriever, Miniature Pinscher, and Lhasa Apso. Thirty-three were neutered males, four were intact males, and 29 were spayed females. The median age of all dogs was 10 years (range 2–15 years).

Dogs in the epithelial neoplasia group ranged from 4 to 15 years of age (median 10 years), and dogs in the mesenchymal neoplasia group ranged from 2 to 12 years of age (median 9 years). The most common presenting complaints were epistaxis (47 dogs, 70%), sneezing (22 dogs, 33%), nasal discharge (11 dogs, 16%), and stertor (eight dogs, 12%). Thirty-two dogs (47.8%) had rhinoscopy-assisted biopsy, 29 (43.3%) had blind nasal biopsy, three (4.5%) had surgical biopsies of the extra-nasal portion of the mass, two (3%) had gross biopsy via necropsy, and one (1.5%) had biopsy obtained via flushing the nasal cavity. Lymph node cytology was performed on 19 dogs (28%). Of these dogs, six (32%) had cytologic changes consistent with metastasis, eight (42%) had nonneoplastic changes (e.g., lymphoid hyperplasia, lymphadenitis), four (21%) were nondiagnostic samples, and one (5%) was consistent with a normal lymph node.

The CT findings for each group are summarized in Table S1. An abbreviated table of CT findings is found in Table 3. The incidence of several CT features differed between dogs with a nasal epithelial tumor and those with a nasal mesenchymal tumor. Dogs with epithelial neoplasia were more likely to show intracranial mass extension ($p$ = 0.04; OR 5.1; 95% CI 1.1–23.9), cribriform plate lysis ($p$ = 0.03; OR 4.5; 95% CI 1.2–15.8), lysis of ipsilateral sphenoid sinus ($p$ < 0.0001; OR 18.7; 95% CI 3.9–85.9), mass extension into the ipsilateral sphenoid sinus ($p$ = 0.01; OR 5.8; 95% CI 1.6–20.2), and fluid in the frontal sinus ($p$ = 0.05; OR 4.7; 95% CI 1.3–16.2) on CT than dogs with mesenchymal neoplasia. Dogs with nasal mesenchymal neoplasia were more likely to show fluid in the ipsilateral maxillary recess ($p$ = 0.01; OR 5.3; 95% CI 1.4–18.6) than dogs with epithelial tumors.

Five dogs were diagnosed with squamous cell carcinoma. Two of these dogs had masses centered in the rostral nasal cavity (specifically the nasal planum) and no associated lysis or mass extension into adjacent structures (Figure 1). The remaining three dogs had masses that were centered on the maxillary or nasal bone,

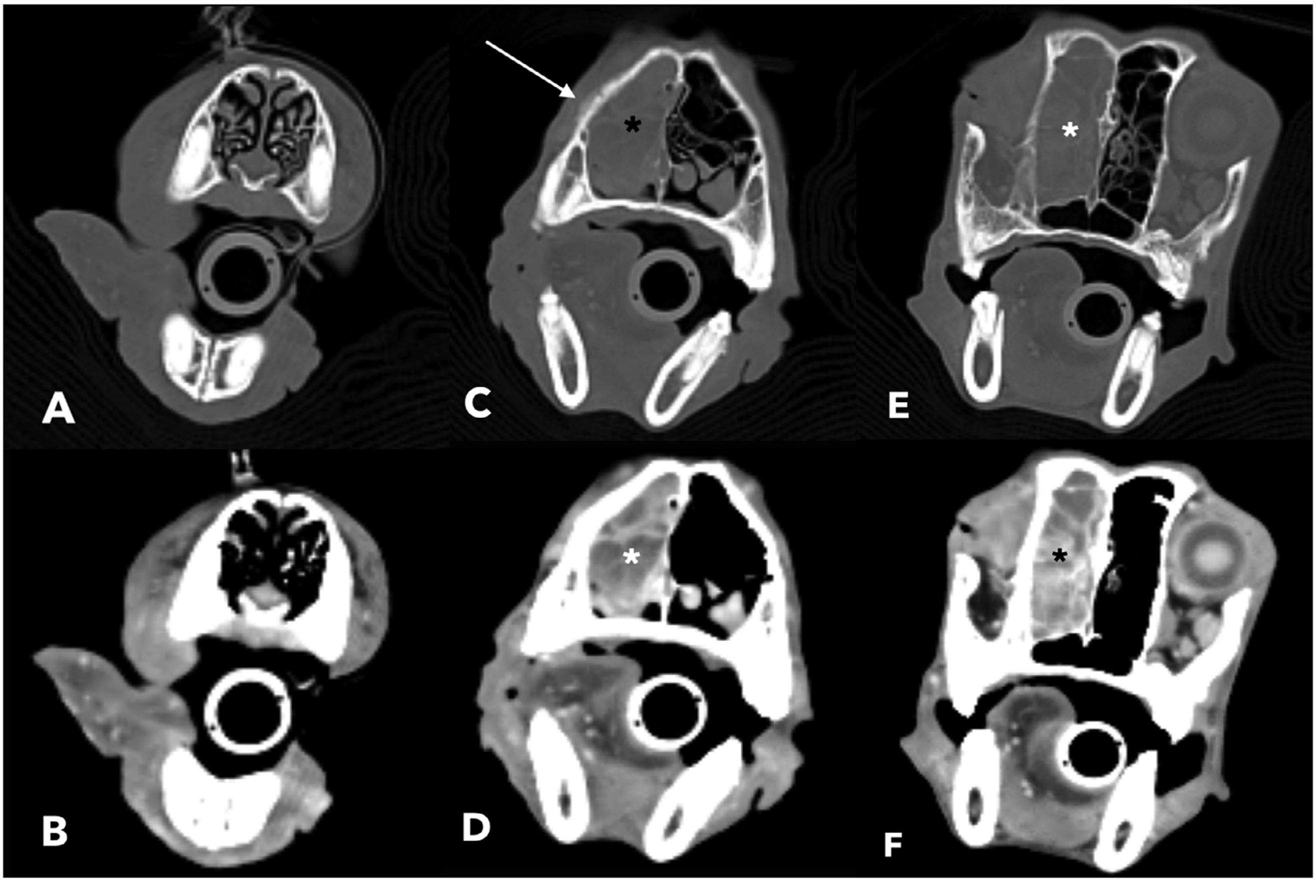


**FIGURE 9** | Transverse, contrast-enhanced, computed tomographic images of a dog diagnosed with osteosarcoma in the nasal cavity. Bone (A, C, E) and soft tissue images (B, D, F) of the rostral (A and B), mid (C and D), and caudal (E and F) aspects of the nasal cavity show moderately contrast-enhancing mass centered in the right caudal nasal cavity, causing moderate lysis of the nasal conchae (asterisks in C and D) and severe lysis of the ethmoidal conchae (asterisks in E and F). Bone images are displayed with a window width of 2700 HU and a window level of 350 HU; soft tissue images are displayed with a window width of 320 HU and a window level of 30 HU. Slice thickness 3 mm.

causing lysis and extending into the nasal cavity and the paranasal soft tissues (Figure 2). These three masses were moderately and heterogeneously enhancing, had periosteal reaction, and caused severe lysis of the adjacent nasal conchae. None of the dogs had mass extension into or lysis of the ipsilateral sphenoid sinus, and only one dog had (minimal) fluid in the ipsilateral sphenoid sinus. Two dogs had fluid in the ipsilateral frontal sinus (one classified as mild, one classified as all). Two dogs had lysis and mass extension into the ipsilateral maxillary recess. New mineralization was present in two dogs—one characterized as moderate and one as mild. One dog had lysis of the ipsilateral orbit (severe), with moderate retrobulbar extension, cribriform plate lysis (mild), and minimal intracranial mass extension (Figure 3).

One of the dogs diagnosed with undifferentiated carcinoma had a mass centered on the nasal planum with no associated lysis or mass extension (Figure 4). This dog had mildly enlarged right mandibular lymph nodes with adjacent mild fat stranding. The cytology of this lymph node was interpreted as metastatic carcinoma with marked reactive hyperplasia.

Of the five dogs diagnosed with osteosarcoma (Figures 5–9), three had masses centered in the mid and two in the caudal nasal cavity. One mass was mildly enhancing, and three were moderately enhancing; one was homogeneously enhancing, and four were heterogeneously enhancing. Two dogs had severe, two dogs had mild, and one dog had no new mineralization in the mass. All dogs had nasal conchal (two severe and three moderate) and ethmoidal conchal (three mild, one moderate, one severe) lysis. Four dogs had periosteal reaction of the adjacent osseous margins (one severe, one moderate, two mild), and the same four dogs had a mass and lysis involving the ipsilateral maxillary recess. All dogs except one had lysis of and mass extension into the ipsilateral maxillary recess; two of these dogs also had fluid in the ipsilateral maxillary recess. None of the dogs diagnosed with osteosarcoma had cribriform plate lysis, a mass in the ipsilateral sphenoid sinus, or intracranial extension.

The three dogs diagnosed with polyps (Figures 10–12) had masses centered in the mid (one dog) or caudal (two dogs) nasal cavity. One mass was mildly enhancing, and two were strongly enhancing, and all masses were heterogeneously enhancing. Two dogs had severe and one dog had mild new mineralization in the mass. All dogs had nasal conchal (two severe and one moderate) and ethmoidal conchal (all moderate), as well as vomer/nasal septal lysis. All dogs had mild nasopharyngeal mass extension. One dog who had mass extension into the ipsilateral frontal sinus also had mass extension into the adjacent soft tissues, causing a mass caudodorsal to the right eye (Figure 11). One dog had cribriform plate lysis (mild) and intracranial extension of the new

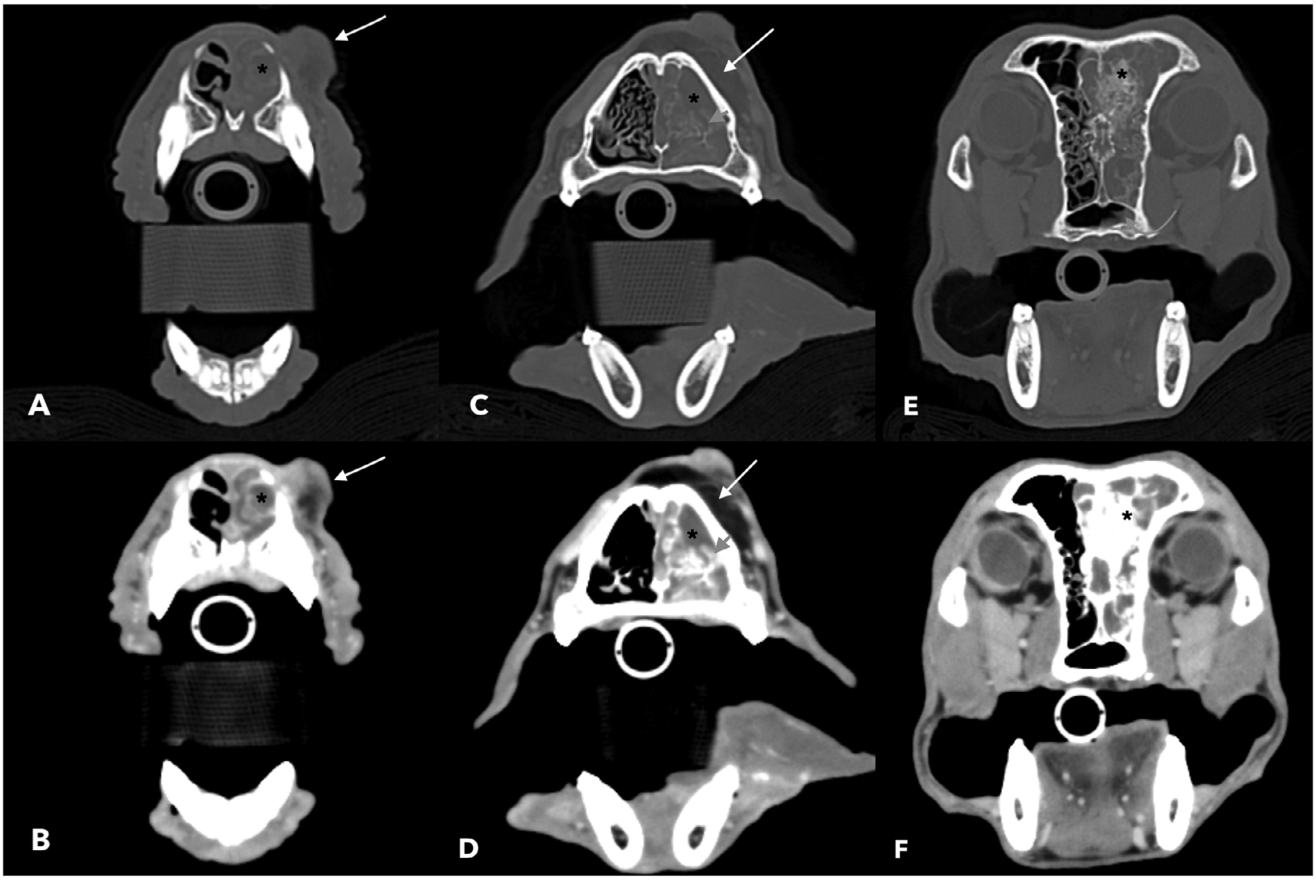


**FIGURE 10** | Transverse, contrast-enhanced, computed tomographic images of a dog diagnosed with a polyp in the nasal cavity. Bone (A, C, E) and soft tissue images (B, D, F) of the rostral (A and B), mid (C and D), and caudal (E and F) aspects of the nasal cavity show a mildly contrast-enhancing mass centered in the left mid nasal cavity, extending caudally, causing moderate conchal lysis (black asterisks in C and D) with severe new mineralization (black asterisks in E and F). This patient had a paranasal soft tissue extension of the mass with a mixed soft tissue and fat attenuation (white arrows in A–D). The rostral left nasal cavity had nonenhancing fluid (black asterisks in A and B). Bone images are displayed with a window width of 2700 HU and a window level of 350 HU; soft tissue images are displayed with a window width of 320 HU and a window level of 30 HU. Slice thickness 2 mm.

mineral (Figure 13). This same dog had a moderate paranasal soft tissue extension of fat-attenuating soft tissue (Figure 10). All dogs had fluid in the ipsilateral frontal sinus, and two dogs had mass extension into the frontal sinus. One dog had lysis of the margins of the ipsilateral frontal sinus. All dogs had fluid in the ipsilateral sphenoid sinus, and two had mass extension into the ipsilateral sphenoid sinus. All dogs had lysis of the ipsilateral maxillary recess; one had fluid and mass extension, one had mass only, and one had fluid only in the ipsilateral maxillary recess.

The CT feature of new mineralization within the mass was present in both epithelial neoplasia (33.3%) and mesenchymal neoplasia (47.4%) groups, with a total of 25 dogs having new mineral. Severe new mineralization was present in six cases (two polyps, two osteosarcoma cases, one fibrosarcoma, and one rhabdomyosarcoma), moderate new mineralization was present in four cases (one undifferentiated sarcoma, two adenocarcinoma, and one squamous cell carcinoma), and mild new mineral was present in 15 cases (two osteosarcoma, one chondrosarcoma, one polyp, one undifferentiated sarcoma, seven adenocarcinoma, one squamous cell carcinoma, and two undifferentiated carcinoma).

## 4 | Discussion

This is the first study describing and comparing CT features of confirmed nasal neoplasia in a large group of dogs to aid in discriminating between epithelial and mesenchymal neoplasia.

Intracranial mass extension (OR 5.1), cribriform plate lysis (OR 4.5), lysis of the ipsilateral sphenoid sinus (OR 18.7), mass extension into the ipsilateral sphenoid sinus (OR 5.8), and fluid in the frontal sinus (OR 4.7) were statistically significantly more likely in dogs with epithelial neoplasia compared to mesenchymal neoplasia. Fluid in the ipsilateral maxillary recess was statistically significantly more likely in dogs with mesenchymal neoplasia (OR 5.3). New mineralization in the mass was not significantly associated with either group, being present in 33.3% of dogs with benign and malignant epithelial neoplasia and 47.4% of dogs with mesenchymal neoplasia, disproving our hypothesis that lesional mineralization would be more common with mesenchymal neoplasia.

The risk of nasal neoplasia is positively correlated with the amount of surface area in the nasal cavity and, thus, with the

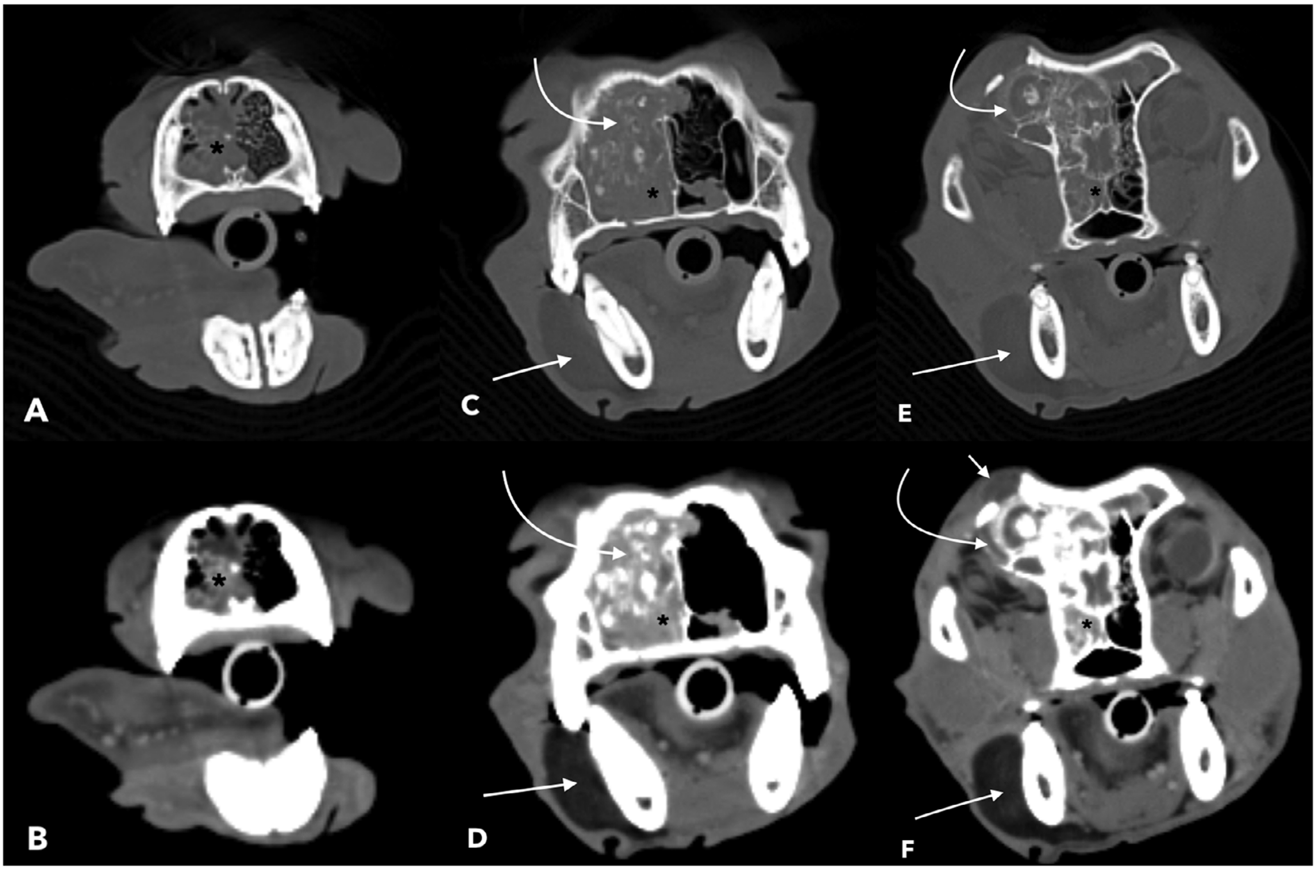


**FIGURE 11** | Transverse, contrast-enhanced, computed tomographic images of a dog diagnosed with a polyp in the nasal cavity. Bone (A, C, E) and soft tissue images (B, D, F) of the rostral (A and B), mid (C and D), and caudal (E and F) aspects of the nasal cavity show a strongly contrast-enhancing mass centered in the right caudal nasal cavity, causing severe conchal and moderate ethmoidal lysis (black asterisks in C–F) with severe new mineral production (curved arrows in C–F). This mass extended into the right retrobulbar space (curved white arrows in E and F) and to the skin surface, causing a mass caudodorsal to the right eye (short white arrow F). This patient also had a fluid in the rostral right nasal cavity (black asterisks in A and B) and a lipoma along the right mandible (long, straight white arrows C–F). Bone images are displayed with a window width of 2700 HU and a window level of 350 HU; soft tissue images are displayed with a window width of 456 HU and a window level of 84 HU. Slice thickness 3 mm.

efficiency of filtering potential environmental carcinogens [10]. Nasal carcinomas more commonly occur in the mid-to-caudal nasal cavity [11], likely due to the increased surface area of the nasal conchae in this region. This could explain why the dogs with epithelial neoplasia were more likely to have lysis and tumor extension into structures located in the caudal nasal cavity (cribriform plate, sphenoid sinus) or into structures caudal to the nasal cavity (cranial cavity) compared to dogs with mesenchymal neoplasia.

Squamous cell carcinoma is the most common neoplasia of the nasal planum [12] and can be locally invasive but rarely metastasizes. Three dogs in our study had neoplasia centered on the nasal planum. Two were diagnosed with squamous cell carcinoma and one with adenosquamous carcinoma. The adenosquamous carcinoma case had lymph node cytology that showed cohesive neoplastic cells consistent with metastatic carcinoma, and some features of the neoplastic cells were suggestive of keratinization, causing a squamous cell origin to be considered. Given that adenosquamous carcinoma has not been reported in the literature to arise from the nasal planum in dogs, and that the distinction between adenosquamous carcinoma, squamous cell carcinoma, and adenocarcinoma with squamous differentiation can be difficult on histopathology [13], it is possible this case was incorrectly classified. The authors feel confident prioritizing squamous cell carcinoma in dogs with nasal tumors that arise from the nasal planum and have no associated lysis.

The other three dogs diagnosed with squamous cell carcinoma had masses centered on the maxillary or nasal bone, causing lysis of these structures and extending into the paranasal soft tissues, rather than being centered on or arising within the nasal cavity, like most other tumors in this study. The nasal cavity of the dog is lined by different types of epithelial cells depending on location, with keratinized stratified squamous epithelium lining the nasal conchae in the vestibular (rostral) portion, pseudostratified squamous epithelial cells in the mid portion, and olfactory epithelium (with basal, sustentacular, and bipolar cells) in the caudal portion [14]. Squamous cell carcinomas can arise from both squamous and non-squamous epithelial tissues [15], since chronically irritated epithelial cells can undergo squamous metaplasia. Therefore, it is reasonable that squamous cell carcinoma tumors could arise from any portion of the nasal cavity. However, it is uncertain why these three tumors were centered on the bony structures when histologically, they arise from the epithelial lining of the nasal conchae. In other studies, squamous cell carcinoma in the oral cavity of cats and in the sinonasal region of horses often presents in a similar way, with the majority of cases having a marked and extensive deep invasion into the bone [16].

The three nasal polyp cases had a large amount of lysis of the intranasal bony structures, and one dog had cribriform plate lysis. These features are more aggressive than nasopharyngeal or middle ear polyps, which typically cause more bony expansion and

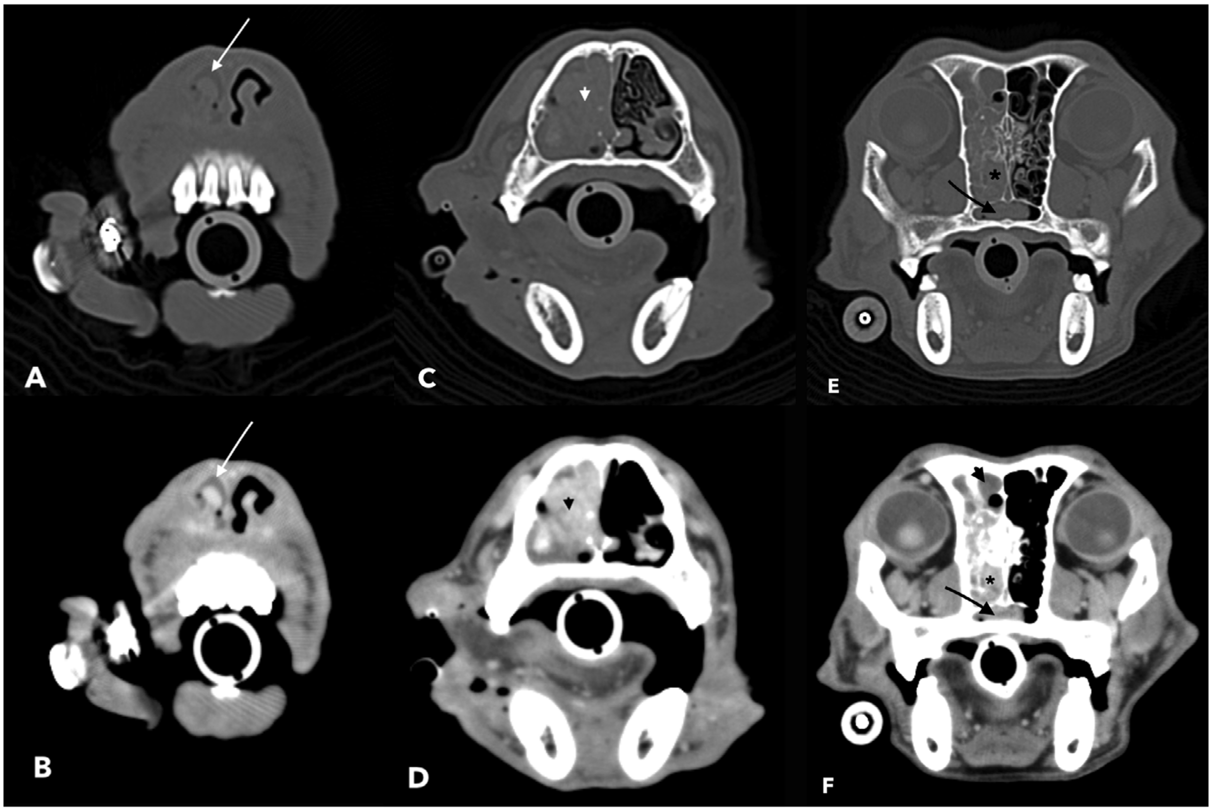


**FIGURE 12** | Transverse, contrast-enhanced, computed tomographic images of a dog diagnosed with a polyp in the nasal cavity. Bone (A, C, E) and soft tissue images (B, D, F) of the rostral (A and B), mid (C and D), and caudal (E and F) aspects of the nasal cavity show a strongly contrast-enhancing mass centered in the right caudal nasal cavity, causing severe conchal (short arrows in C and D) and moderate ethmoidal (black asterisks in E and F) lysis with mild nasopharyngeal mass extension (long black arrows in E and F). Fluid extended rostrally (white arrows and A and B) and caudally into the frontal sinus (short black arrow in F). Bone images are displayed with a window width of 2700 HU and a window level of 350 HU; soft tissue images are displayed with a window width of 320 HU and a window level of 30 HU. Slice thickness 2 mm.

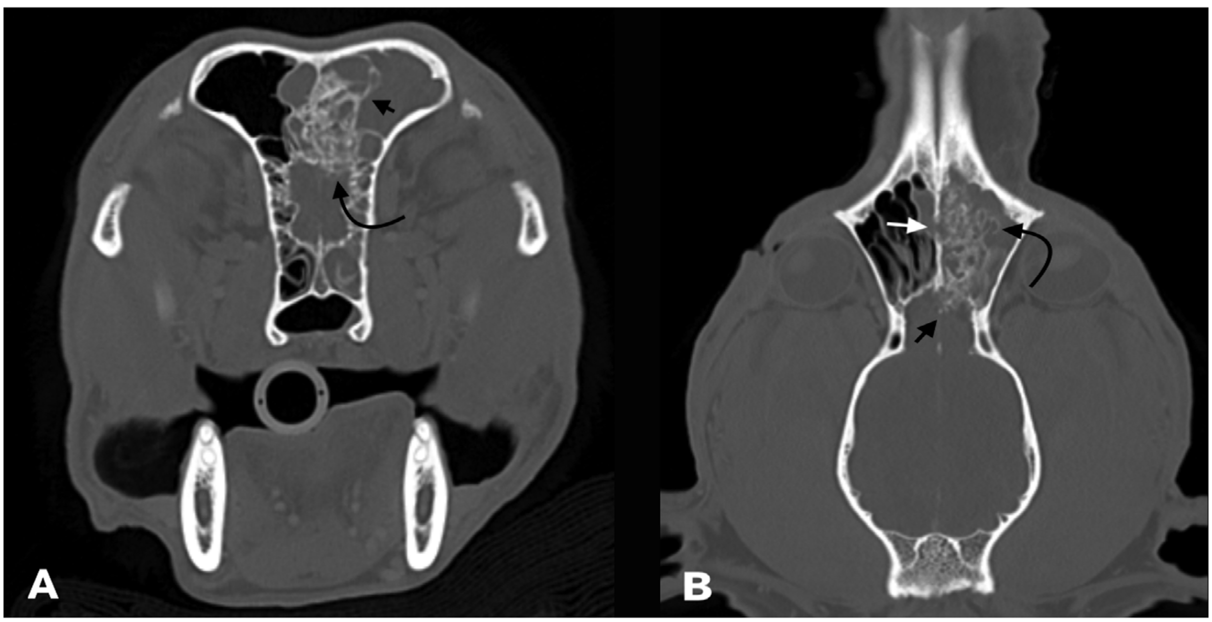


**FIGURE 13** | Legend on next page.

**FIGURE 13** | Transverse (A) and dorsal (B) contrast-enhanced computed tomographic images of one of the dogs diagnosed with nasal cavity polyp (same patient as in Figure 10) show severe new mineral formation (curved arrow in B and short black arrow in A), mild cribriform plate lysis (curved arrow in A), and mild intracranial extension of the new mineral. This patient also has mild nasal septum lysis (short white arrow in B). Computed tomographic images were reconstructed using a bone algorithm with a 2-mm slice thickness and are displayed with a window width of 2700 HU and a window level of 350 HU.

pressure necrosis-related lysis [17]. From a prior study [18] of dogs with nasal polyposis, all those that had CT imaging (four out of 13) had total or partial turbinate lysis, and most (three out of four) had nasal septal or nasal bone lysis. These findings are similar to the findings in the current study. Previous literature [19] has identified that nasal polyps frequently occur in conjunction with nasal carcinoma, with 21.4% of nasal carcinomas concurrently having nasal polyps. Also in this study, of the 50 cases originally diagnosed with nasal polyps, 26 were re-biopsied, and in 35% of these dogs, the second biopsy revealed a malignancy. This raises the possibility that the cases with a histologic diagnosis of polyp in the current study did not have representative biopsies.

The CT finding of new mineralization within the mass was not statistically associated with either tumor group and was present in several different types of benign and malignant neoplasia. The cases with severe new mineral were either mesenchymal tumors (osteosarcoma, fibrosarcoma, or rhabdomyosarcoma) or polyps. The etiology of this new mineral is not definitively known; however, tumor-induced osteoblastic activity or dystrophic mineralization from chronic inflammation are both possible. Although four out of five osteosarcoma cases had new mineral, one did not; therefore, the lack of new mineral formation does not rule out osteosarcoma.

The main limitation of this study is the retrospective design. Additionally, the authors were aware of the diagnosis of neoplasia, which may have biased feature recognition. Also, due to a small number of cases in each specific type of neoplasia (especially polyps), statistical analysis between specific types of neoplasia was not possible.

In conclusion, in dogs diagnosed with nasal tumors, an epithelial neoplasm should be prioritized if intracranial mass extension, cribriform plate lysis, or lysis/mass extension into the ipsilateral sphenoid sinus is present. A diagnosis of squamous cell carcinoma should be considered (1) if the mass is centered on the nasal planum with no associated lysis or (2) if the mass is centered on the maxillary or nasal bone, causing lysis and extending into the paranasal soft tissues. The presence of new mineralization in the mass does not help prioritize a specific tumor type.

## Author Contributions

Conception and design: Stacy Cooley, Stacie Summers, Kelley Van Scoyk, and Lauren Newsom. Acquisition of data: Stacy Cooley and Lauren Newsom. Analysis and interpretation of data: Kelley Van Scoyk, Stacie Summers, Stacy Cooley, and Lauren Newsom. Drafting of the article: Kelley Van Scoyk. Reviewing article for intellectual content: Kelley Van Scoyk, Stacie Summers, Stacy Cooley, and Lauren Newsom. Final approval of the completed article: Kelley Van Scoyk, Stacie Summers, Stacy Cooley, and Lauren Newsom. Agreement to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved: Kelley Van Scoyk, Stacie Summers, Stacy Cooley, and Lauren Newsom.

## Disclosure

The authors have nothing to report.

## Conflicts of Interest

The authors declare no conflicts of Interest.

## Data Availability Statement

Supporting Information can be found.

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## Supporting Information

Additional supporting information can be found online in the Supporting Information section.

**Supporting File 1**: vru70099-sup-0001-TableS1.docx